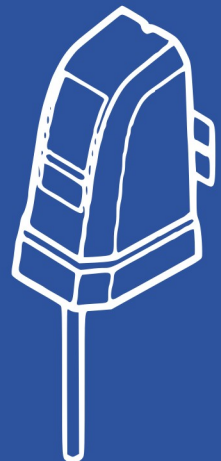
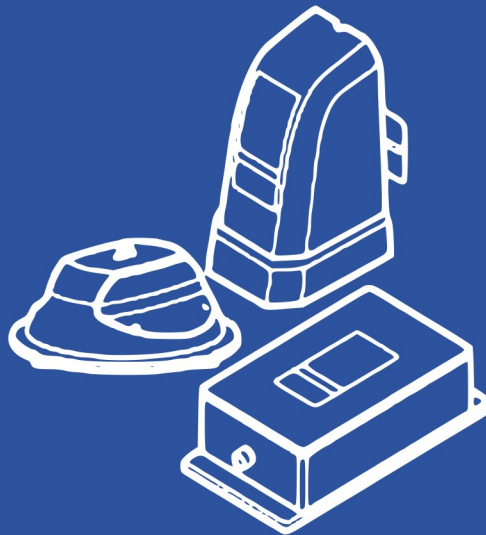
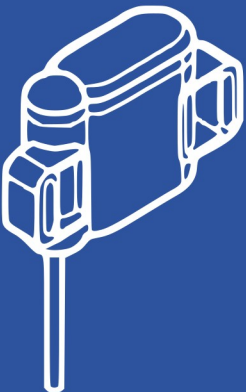




# LIDoTT Alarm 2 4G LTE Cat 1 Manual

Installation, Configuration and User Guide | V0.0.1 | September 2026



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## Document Control and Conventions

| Item                   | Detail                                                                  |
|------------------------|-------------------------------------------------------------------------|
| Document               | LIDoTT ALARM 2 4G LTE Cat 1 Installation, Configuration and User Manual |
| Product                | Detectronic LIDoTT ALARM 2, 4G LTE Cat 1 variant                        |
| Version                | V0.0.1                                                                  |
| Issue date             | September 2026                                                          |
| Configuration software | LIDoTT Configurator (Android or Windows), latest version                |

### Revision History

| Version | Date           | Description                                                     |
|---------|----------------|-----------------------------------------------------------------|
| V0.0.1  | September 2026 | First issue for the 4G LTE Cat 1 variant of the LIDoTT ALARM 2. |

### Conventions Used in This Manual

Important information is highlighted in coloured panels. Read every WARNING and CAUTION before starting work.

**WARNING** Risk of serious injury, or of compromising the hazardous area protection of the equipment.

**CAUTION** Risk of damage to the equipment, loss of data or unreliable operation.

**NOTE** Additional information that helps you complete a task correctly.

**TIP** Good practice from field experience that saves time on site.

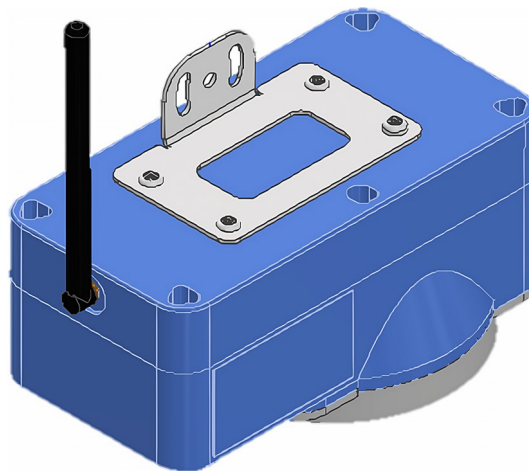
Buttons and screen names in the LIDoTT Configurator are shown in bold blue, for example **‘REFRESH DEVICES’**. Field names are shown in bold, for example **Empty Distance**.

## 1.0 Introduction

---

The Detectronic LIDoTT ALARM 2 4G LTE Cat 1 is a compact, battery powered monitoring device designed to provide early warning of high level events to support sewer network monitoring, flood forecasting and environmental data collection.

It is specifically engineered for challenging underground environments, such as maintenance holes and chambers, where dependable and low maintenance monitoring is essential. Its uncomplicated design and long battery life make it ideal for rapid deployment, temporary monitoring and targeted alarm only applications.



**Figure 1:** LIDoTT ALARM 2 4G LTE Cat 1 with standard whip antenna

### How the ALARM 2 Works

- 1. Measure.** A downward facing radar measures the distance from the skirt of the ALARM 2 to the water surface at the selected sample rate.
- 2. Calculate.** The water level is calculated as Empty Distance minus the sensor reading.
- 3. Evaluate.** Each reading is compared with up to five alarm levels, using Threshold, Hysteresis and Persistence to avoid false alarms.
- 4. Report.** The ALARM 2 dials out over the public 4G LTE network to the monitoring platform at the send interval (or the alarming send interval when in alarm) and sends a daily heartbeat.

Because the LTE Cat 1 modem connects straight to the mobile carrier's network, no LoRaWAN gateway or network server is required. LTE Cat 1 operates on standard 4G LTE networks and does not rely on dedicated low power IoT network coverage. The ALARM 2 supports the 4G LTE bands used by Australian carriers (Bands 1, 3, 5, 7, 8 and 28), including the 700 MHz Band 28, which gives the best coverage into below ground chambers.

## Key Features

- Radar level measurement from 0 to 20 m with no dead band.
- Up to five alarm levels, each with its own Threshold, Hysteresis and Persistence.
- Separate normal and alarming sample rates and send intervals.
- Movement and fall alert if the device is disturbed in the chamber.
- Daily heartbeat with battery voltage, water level, temperature, alarm status and angle on all three axes.
- Configuration by Bluetooth using the LIDoTT Configurator, with remote configuration and firmware updates over 4G.
- Removable boom arm, so the device can be lifted clear for jetting or other works without damage.
- ATEX, IECEx and UKEX certified for zones 0, 1 and 2.

This manual is intended for trained installers, operational staff and engineers responsible for deploying and managing monitoring equipment within the wastewater network. By following it, users can ensure the ALARM 2 is installed correctly, operates reliably and delivers accurate and timely alarm notifications.

## 2.0 Safety

---

**WARNING** Maintenance holes are confined spaces. Entry must only be made by trained personnel under the site Safe Work Method Statement (SWMS), a confined space entry permit and continuous atmospheric monitoring, in accordance with the WHS Regulations and AS 2865. Install from the surface wherever practical.

### Hazardous Area Certification

The LIDoTT ALARM 2 is **ATEX, IECEx and UKEX certified** for use within hazardous areas. It is assumed that all necessary system design, specification and engineering factors have been considered by suitably competent personnel before installation begins.

- The equipment may be used in **zones 0, 1 and 2** with flammable gases and vapours of apparatus groups IIA and IIB, and temperature classes T1, T2, T3 and T4.
- The equipment is only certified for use in ambient temperatures from **-20 °C to +60 °C** and must not be used outside this range.
- Installation shall be carried out by suitably trained personnel in accordance with the applicable code of practice. In Australia this is AS/NZS 60079.14, together with the conditions of the IECEx certificate.
- There is no special checking or maintenance other than a health check (see 9.0 Maintenance). Regarding explosion safety, it is not necessary to check for correct operation.
- The equipment contains no user replaceable parts and must not be repaired by the user. Repairs shall only be carried out by the manufacturer or its approved agents, in accordance with the applicable code of practice.
- Only fit the 4G LTE antenna types supplied or approved by Detectronic for this variant.
- Where the equipment may encounter aggressive substances, such as acidic liquids or gases that may attack metals, or solvents that may affect polymeric materials, the user must take suitable precautions so that the type of protection is not compromised.

### Special Conditions of Use (certificate suffix X)

The certificate number has an 'X' suffix, which indicates that special conditions of installation and use apply. Those installing or inspecting this equipment must have access to the certificate or these instructions. The conditions are reproduced below.

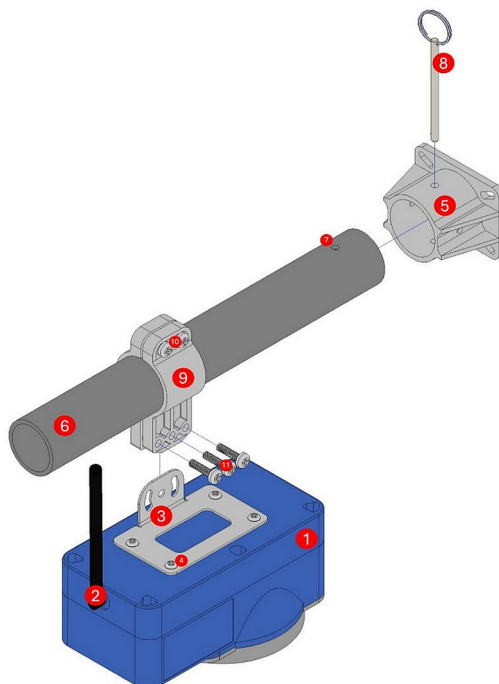
**WARNING** Under certain extreme circumstances, the non metallic parts of the enclosure may generate an ignition capable level of electrostatic charge. Do not install the equipment where external conditions are conducive to the build up of electrostatic charge on such surfaces. Clean the equipment with a damp cloth only. This is particularly important in a zone location.

Due to Detectronic's policy of continuous product development, products may differ in detail from the information in this manual. Other versions of this manual are available from the login area of the Detectronic website or from a Detectronic representative. Detectronic Ltd accepts no responsibility for any loss or damage caused directly or indirectly by using software or equipment it supplies other than in accordance with this manual.

## 3.0 Specifications and Components

| Parameter                  | Specification                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Measurement</b>         | Radar distance measurement to the water surface                                                             |
| <b>Range</b>               | 0 to 20 m, no dead band                                                                                     |
| <b>Level calculation</b>   | Water Level (mm) = Empty Distance - Sensor Reading                                                          |
| <b>Communications</b>      | 4G LTE Cat 1, Australian Bands 1, 3, 5, 7, 8 and 28                                                         |
| <b>Antenna</b>             | SMA connector; standard whip, or enhanced external 4G LTE antenna where coverage requires                   |
| <b>SIM</b>                 | Fitted and activated at supply; not user replaceable                                                        |
| <b>Local configuration</b> | Bluetooth, using the LIDoTT Configurator on Android or Windows                                              |
| <b>Remote management</b>   | Configuration and firmware updates over the 4G connection                                                   |
| <b>Sample rate</b>         | Disabled, 1, 2, 5 or 15 minutes, with a separate alarming sample rate                                       |
| <b>Send interval</b>       | Disabled, 15 or 30 minutes, or 1, 2, 3, 4, 6, 8, 12 or 24 hours, with a separate alarming send interval     |
| <b>Alarms</b>              | Up to 5 levels, each with Threshold, Hysteresis and Persistence, plus movement and fall alert               |
| <b>Daily heartbeat</b>     | Battery voltage, water level, temperature, alarm status and angle on 3 axes                                 |
| <b>Power</b>               | Internal battery; life depends on sample rate, send interval and signal strength                            |
| <b>Hazardous area</b>      | ATEX, IECEx and UKEX; zones 0, 1 and 2; groups IIA and IIB; T1 to T4                                        |
| <b>Ambient temperature</b> | -20 °C to +60 °C                                                                                            |
| <b>Mounting</b>            | Removable boom arm with slide, twist and tilt adjustment; STANAG 4694 accessory rail for the alignment tool |

## Components



**Figure 2:** ALARM 2 and boom arm components

| No. | Component                        |
|-----|----------------------------------|
| 1   | LIDoTT ALARM 2 4G LTE Cat 1      |
| 2   | 4G LTE Antenna                   |
| 3   | Metal Alarm Mounting Socket      |
| 4   | Four M4 x 8 mm Bolts             |
| 5   | Boom Arm Wall Mount              |
| 6   | Boom Arm Pipe                    |
| 7   | Hole for Locking Pin             |
| 8   | Locking Pin                      |
| 9   | Boom Arm to Alarm Device Bracket |
| 10  | Two M5 x 20 mm Bolts             |
| 11  | Three M5 x 20 mm Bolts           |

## Tools and Equipment

The following tools and consumables are recommended for installation and maintenance:

| Installation tools                                            | Consumables, devices and information                                    |
|---------------------------------------------------------------|-------------------------------------------------------------------------|
| Small screwdriver for 3 mm slot head screws, plus cross head  | Plastic cable ties: 20 at 100 mm x 2.5 mm, and 5 at 300 mm x 4.6 mm     |
| Medium screwdriver for 4 mm slot head screws, plus cross head | Roll of strong adhesive tape, approximately 50 mm wide                  |
| Allen key 5 mm ("T" bar type)                                 | Innotec High Temp Wax Spray                                             |
| 10 mm spanner                                                 | Countersunk screws and split pins                                       |
| Ratchet and spanner with 17 mm socket                         | Android smart phone or Windows PC with the latest LIDoTT Configurator   |
| Drill with 5 mm bit, and bits to suit the wall fixings        | Mobile phone or 4G LTE signal tester on the same carrier as the ALARM 2 |
| Snips (wire cutters) and locking pliers (vice grips)          | SIM and platform details (see 4.0 Planning and Preparation)             |
| Magnet (to wake the ALARM 2)                                  | Digital camera for site records                                         |
| Laser alignment tool and spirit level                         | Three Digit Cut Cat 5 gloves and site PPE                               |

## 4.0 Planning and Preparation

Good preparation is the key to a fast, right first time installation. Follow the workflow below and collect the site information before leaving the depot.

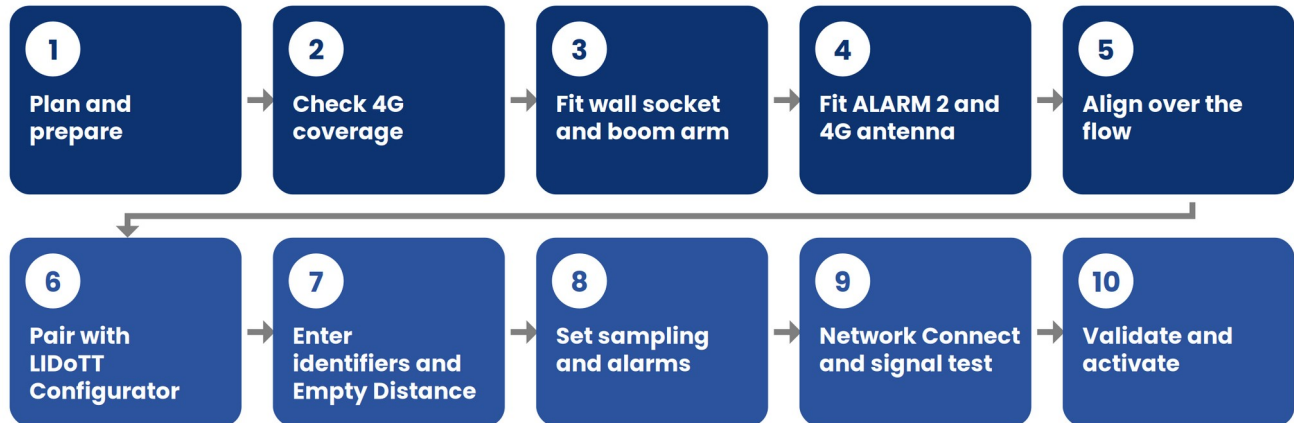


Figure 3: Installation and commissioning workflow

### Information Required Before Installation

| Information                                  | Purpose                                        | Source                                |
|----------------------------------------------|------------------------------------------------|---------------------------------------|
| Manhole Reference and Site Name              | Identify the device on the monitoring platform | Customer                              |
| Alarm levels, hysteresis and persistence     | Set up to 5 alarms                             | Customer requirements                 |
| Sample rates and send intervals              | Balance reporting against battery life         | Customer requirements                 |
| ICCID, APN and any APN username and password | Connect to the 4G network                      | SIM provider                          |
| Server address and port                      | Send data to the monitoring platform           | Detectronic or platform administrator |
| SWMS, permits and gas monitoring             | Safe access to the maintenance hole            | Site owner and employer               |

### Above Ground Coverage Check

Before installing any equipment, confirm that 4G LTE coverage is available at the maintenance hole.

1. Fit the tester or phone with a SIM from the same carrier as the ALARM 2 and set it to 4G (LTE) only.
2. With the maintenance hole cover open, take readings at surface level and just inside the chamber opening, allowing each reading to settle.
3. Record RSRP, RSRQ, SINR, the network operator and the LTE band, and photograph the readings for the site record.

4. Compare the readings with the Signal Strength Guide in section 6.0.

**NOTE** Signal inside the chamber will be lower than at the surface, and lower again once the cover is closed. Aim for Good or better at the chamber opening. Where the signal is Poor, plan to fit an enhanced external 4G LTE antenna.

## Boom Arm Preparation

1. Measure the required length of the boom arm to suit the chamber's dimensions.
2. Cut the boom arm to length and insert it into the wall socket, with the end of the arm flush with the back of the bracket.
3. Drill a 5 mm hole straight through the boom arm, through the locking pin holes on the sides of the mounting socket.
4. Check that the locking pin passes through the boom arm and mounting socket and locks in place. The fit must be tight to prevent disconnection.
5. Separate the arm, locking pin and mounting socket to continue with the installation.

## 5.0 Installation

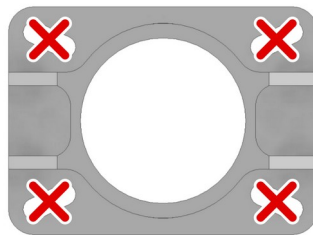
Detectronic recommend installing the LIDoTT ALARM 2 with the boom arm for optimum performance. The boom arm allows several types of mounting and can be removed with the device for jetting or other works in the chamber.

### Wall Socket Installation

1. Identify a location for the socket so that the device will sit over the centre of the flow. The wall must be sturdy enough to support the socket, boom arm, device and antenna.

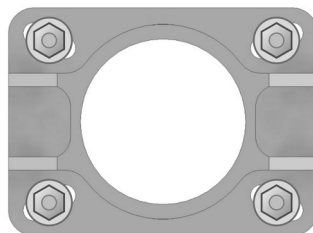
**CAUTION** Leave enough clearance for the antenna, which must be mounted vertically on the ALARM

2. Keep the antenna clear of metal surfaces and as close to the cover as practical.



**Figure 4:** Mark the four fixing holes through the slots in the socket

2. Hold the boom arm socket against the wall and mark the four holes for drilling, ensuring the socket is the correct way up.
3. Remove the socket and drill four holes to a depth suitable for your chosen fixings.
4. Align the socket, insert the fixing bolts and semi tighten. Before inserting one of the bolts, attach the locking pin lanyard to it so the pin cannot fall into the chamber.

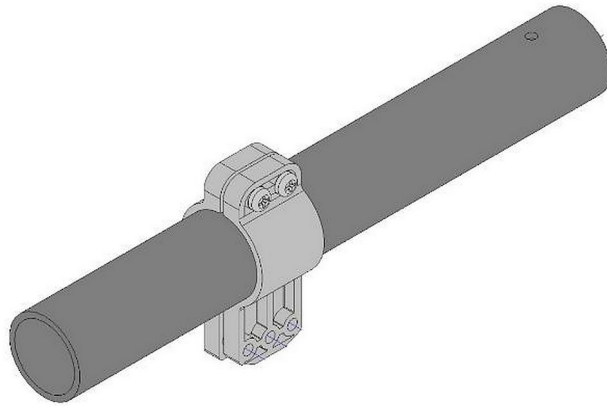


**Figure 5:** Socket fixed to the wall

5. Check the socket alignment and fully tighten the fixing bolts.

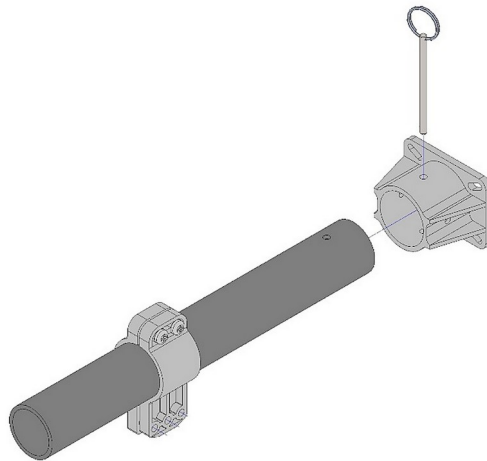
## Boom Arm

1. If using an antenna bracket, place it on the boom arm and semi tighten with two of the M5 x 20 mm bolts.
2. Fit the device bracket to the boom arm and semi tighten with two of the M5 x 20 mm bolts.



**Figure 6:** Device bracket fitted to the boom arm

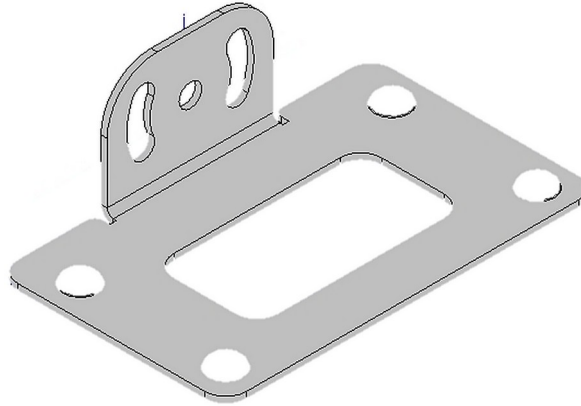
3. Insert the boom arm into the socket and secure it with the locking pin, with the arm over the middle of the flow.



**Figure 7:** Boom arm secured in the wall socket with the locking pin

## ALARM 2 Bracket and Antenna Connection

1. Fit the metal mounting socket to the top of the ALARM 2 using **four M4 x 8 mm** bolts.

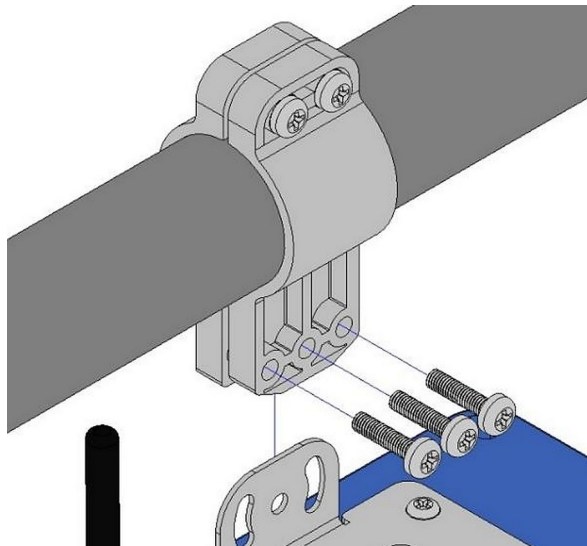


**Figure 8:** Metal alarm mounting socket

2. Screw the 4G LTE antenna onto the SMA connector on the side of the ALARM 2, next to the activation spot (see Pairing & Disconnecting). Tighten by hand only.

**CAUTION** Only fit the 4G LTE antenna supplied for this variant. LoRaWAN antennas are tuned to a different frequency range and will severely reduce 4G performance.

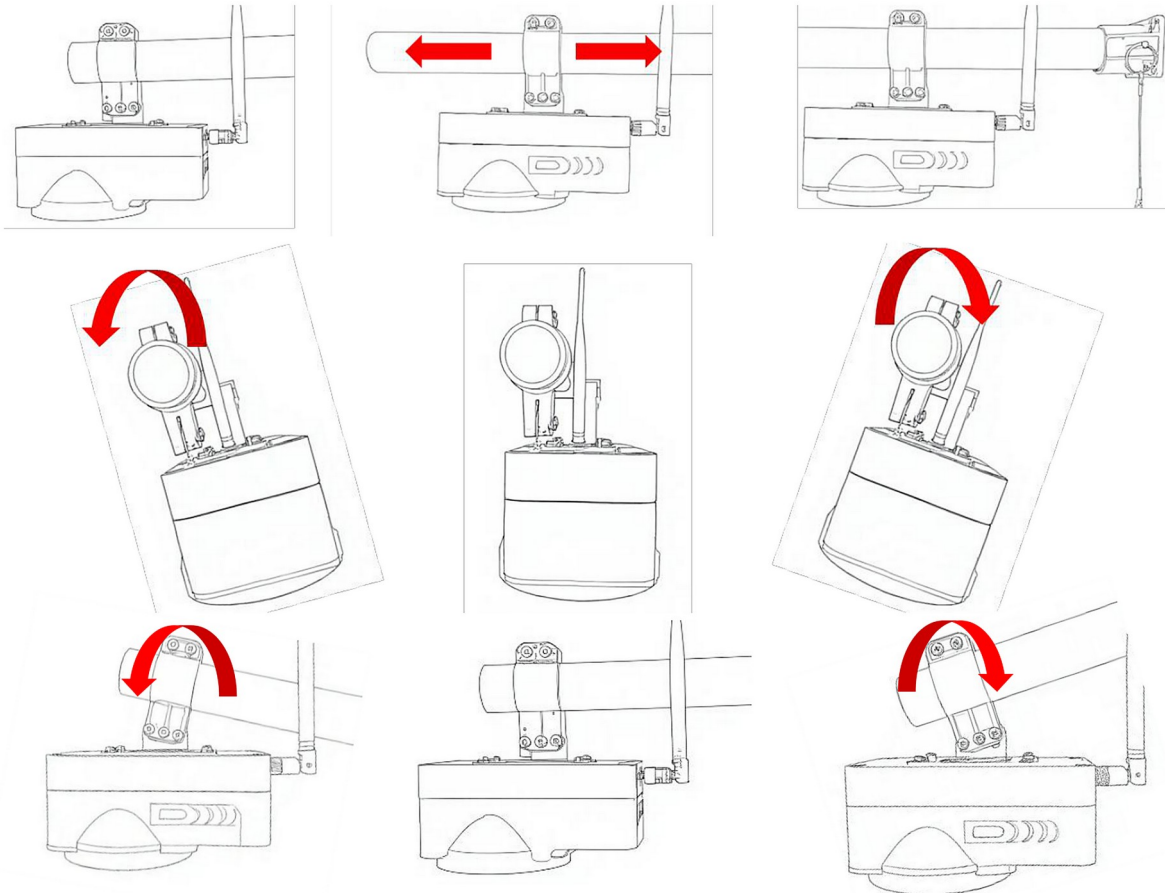
3. Offer the device up to the boom bracket, fit **one** M5 x 20 mm bolt and semi tighten, then fit and semi tighten the remaining **two** M5 x 20 mm bolts.



**Figure 9:** ALARM 2 fitted to the boom arm bracket

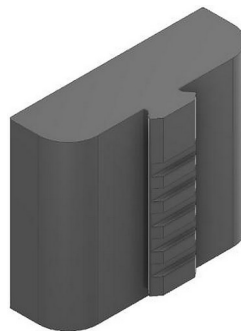
## Boom Arm Adjustments and Alignment

The boom arm offers slide, twist and tilt adjustment. Slightly loosen the device bracket bolts, then slide, twist or tilt the ALARM 2 into position so that the radar points straight down at a smooth section of flow, clear of ladders, steps, benching edges and other obstructions. Fully tighten all bolts once aligned.



**Figure 10:** Slide (top), twist (middle) and tilt (bottom) adjustments

Detectronic offers a laser alignment tool to help position the ALARM 2. Attach the tool to the side and top of the ALARM 2 and switch on the laser to see where the radar is pointing. The side mounted tool is offset 80 mm from the centre of the radar. The Configurator also has a built in alignment tool (see 6.0).



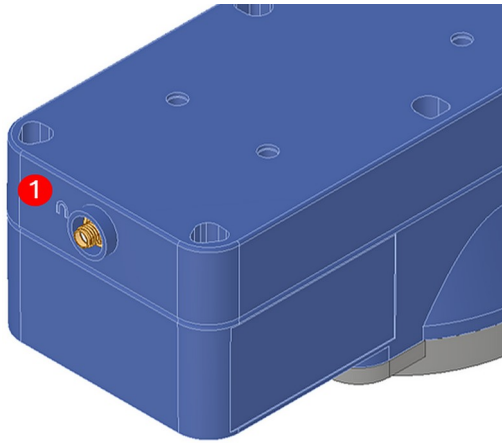
**Figure 11:** Laser alignment tool (laser not fitted), STANAG 4694 NATO rail mounting

## 6.0 Setup and Configuration

---

The ALARM 2 is supplied with default settings and is configured during installation using the **LIDoTT Configurator**. Use the **'PREV'** and **'NEXT'** buttons at the bottom of the screen to move between installation screens. **'PREV'** on the first screen, or **'NEXT'** on the Validate Details screen, returns to the home screen.

### Pairing & Disconnecting



**Figure 12:** Activation spot (1) next to the SMA antenna connector

1. Wake the ALARM 2 by holding a magnet on the activation spot for 5 to 10 seconds.
2. Open the LIDoTT Configurator and tap **'REFRESH DEVICES'**. Wait 5 to 10 seconds for the device to advertise itself.
3. Tap the ALARM 2's serial number to start initialising, for example **"ALM\_007\_1462600000001"**.
4. When initialising is complete, the device is connected to the LIDoTT Configurator.

To disconnect, tap the menu icon then tap **'Disconnect'**.

## Home Screen

| No. | Home screen field           |
|-----|-----------------------------|
| 1   | Serial Number               |
| 2   | IMEI                        |
| 3   | Site Name                   |
| 4   | Firmware Version            |
| 5   | Last Attempted Dial Out     |
| 6   | Last Successful Dial Out    |
| 7   | Last Received Communication |
| 8   | Device Status               |
| 9   | Commissioning Status        |

## Firmware Update

If the firmware is out of date when you first pair, you will be prompted to update to the latest version supplied with the Configurator. Firmware can also be updated remotely over the 4G connection from the monitoring platform.

1. Tap the menu icon  on the Configurator.
2. Tap **'Diagnostics'**, then **'Update Firmware'**. The current firmware version is shown, with an advisory symbol if a newer version is available.
3. Tap **'Latest Version'** to install the firmware supplied with the Configurator, or **'From File'** to select a firmware file on your device.
4. Wait for the *"Update Complete"* message. The ALARM 2 restarts and disconnects; pair again to continue.

**NOTE** Keep the LIDoTT Configurator up to date so that the latest firmware is always available on site. Stay within Bluetooth range until the update is complete.

## Configuration Install

### Device Identifiers

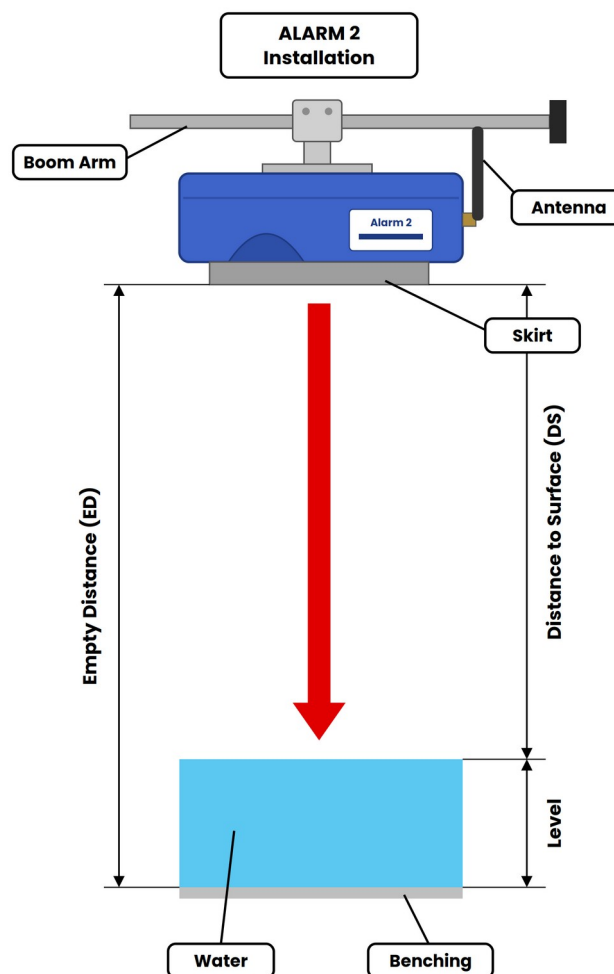
1. **Manhole Reference:** the unique identifier of the maintenance hole where the ALARM 2 is installed.
2. **Site Name:** the name of the site.
3. **Serial Number:** the unique identifier of the ALARM 2, populated automatically.

## Alignment

1. **Built in Alignment Tool:** shows the tilt of the ALARM 2 to help level it during installation.
2. **Scale of Alignment Tool:** enlarges the alignment tool on screen.

## Channels and Empty Distance

1. **Battery Reading:** displays the current battery reading when '**REFRESH**' is tapped.
2. **Level Reading:** displays a level reading when '**REFRESH**' is tapped. Tap '**x5**' for five readings.
3. **Empty Distance:** enter the distance in mm from the bottom of the ALARM 2 skirt to the bottom of the channel (see Figure below). Enter this before taking level readings.
4. **Sample Rate (Mins):** how often the ALARM 2 takes a level reading: Disabled, 1, 2, 5 or 15 minutes.
5. **Alarming Sample Rate:** how often readings are taken whilst in an alarm state. To keep the same rate, mirror the Sample Rate.
6. **Level Readings Graph:** displays the level readings requested manually during installation.

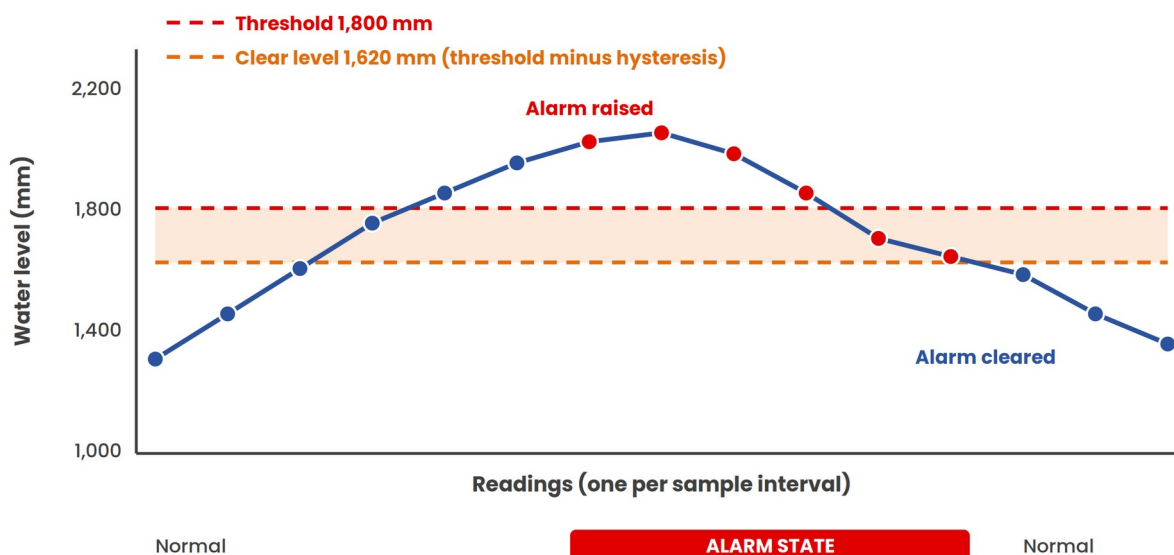


**Figure 13:** Empty Distance, Distance to Surface and Level

**TIP** Worked example: Empty Distance is 2,400 mm and the sensor reads 1,650 mm to the surface.  
Level = 2,400 - 1,650 = **750 mm**. Always confirm the first level reading against a manual measurement.

## Alarms

1. **Send Interval:** how often, at most, the ALARM 2 dials out over 4G to send levels to the platform: Disabled, 15 or 30 minutes, or 1, 2, 3, 4, 6, 8, 12 or 24 hours.
2. **Alarming Send Interval:** how often, at most, the ALARM 2 dials out whilst in an alarm state. The same options apply.
3. **Alarms:** toggle Alarms 1 to 5. Alarm 1 is on by default. Each alarm has its own Threshold, Hysteresis and Persistence.
4. **Threshold (mm):** the water level that must be reached for the alarm to be raised. This is set by customer requirements and is typically 70 to 75% of the Empty Distance.
5. **Hysteresis (mm):** the distance the level must fall below the threshold before the alarm clears. It stops the alarm switching on and off when the level hovers near the threshold. Around 10% of the threshold is a common starting point.
6. **Persistence:** the number of further consecutive readings above the threshold, after the first, needed to raise the alarm.



**Figure 14:** How Threshold, Hysteresis and Persistence work together (Persistence = 2)

**TIP** Worked example: Empty Distance 2,400 mm. Threshold at 75% = **1,800 mm**. Hysteresis at 10% of threshold = **180 mm**, so the alarm clears below 1,620 mm. Persistence 2: the alarm is raised on the third consecutive reading at or above 1,800 mm.

**CAUTION** Every dial out powers the 4G modem and registers it on the mobile network, which uses far more energy than a level reading. Choose the longest send intervals that meet the customer's reporting needs. Poor signal also shortens battery life, because the modem stays on longer and retries more often.

## 4G LTE Network

The ALARM 2 is supplied with its SIM fitted and activated. The SIM is not user replaceable and the enclosure must not be opened; contact Detectronic if the SIM or carrier needs to change.

1. **IMEI:** International Mobile Equipment Identity, the unique 15 digit identifier of the 4G modem, used to register the device on the monitoring platform.
2. **ICCID:** Integrated Circuit Card Identifier, the unique serial number of the SIM, used by the SIM provider to activate and manage the service.
3. **APN:** Access Point Name. Enter it exactly as supplied by the SIM provider; this may be a public APN or a private APN for utility managed SIMs.
4. **APN Username and Password:** only required for some private APNs. Leave blank unless supplied.
5. **Server Address and Port:** the host name or IP address, and port, of the monitoring platform.
6. **'NETWORK CONNECT':** powers the 4G modem, registers on the mobile network and completes a test dial out to the server.
7. **Connection:** a ✓ confirms 4G registration and a successful dial out to the server.
8. **Signal Strength:** shows RSRP, RSRQ and SINR, with the network operator and LTE band.

## Signal Strength Guide

Run the signal test with the ALARM 2 in its installed position and compare the results with the table below. Aim for Good or better with the cover open, as the signal will fall once the cover is closed.

| Rating           | RSRP                | SINR             | Expected performance                              |
|------------------|---------------------|------------------|---------------------------------------------------|
| <b>Excellent</b> | -80 dBm or better   | 20 dB or better  | Very strong signal, reliable dial outs            |
| <b>Good</b>      | -80 to -90 dBm      | 13 to 20 dB      | Strong signal, reliable dial outs                 |
| <b>Baseline</b>  | -90 to -100 dBm     | 0 to 13 dB       | Acceptable signal, occasional retries             |
| <b>Poor</b>      | -100 to -110 dBm    | -3 to 0 dB       | Weak signal, frequent retries, higher battery use |
| <b>Very Poor</b> | Worse than -110 dBm | Worse than -3 dB | Very weak signal, dial outs likely to fail        |

RSRP worse than -100 dBm or SINR worse than 0 dB in the installed position requires an enhanced external 4G LTE antenna. RSSI alone is not a reliable measure of 4G link quality because it includes noise and interference from neighbouring cells; always use RSRP and SINR.

## Validate Details

This screen summarises the setup: Serial Number, IMEI, Manhole Reference, Site Name, 4G Network Status, Coordinates and Device Status. Check every item, then tap **'NEXT'** to complete the installation and change the device from standby to active.

## 7.0 Commissioning Checklist

Work through this checklist before leaving site. Keep a copy with the site record and photographs.

| Check                    |                                                                               |
|--------------------------|-------------------------------------------------------------------------------|
| <input type="checkbox"/> | SWMS followed, permits in place and atmosphere monitored throughout           |
| <input type="checkbox"/> | Above ground 4G coverage recorded (RSRP, RSRQ, SINR, operator and band)       |
| <input type="checkbox"/> | Wall socket fixings fully tightened; locking pin fitted with lanyard attached |
| <input type="checkbox"/> | Boom arm over the centre of the flow; all bracket bolts fully tightened       |
| <input type="checkbox"/> | 4G LTE antenna fitted vertically, hand tight and clear of metal               |
| <input type="checkbox"/> | ALARM 2 level, and radar clear of ladders, steps and other obstructions       |
| <input type="checkbox"/> | Firmware updated to the latest version                                        |
| <input type="checkbox"/> | Manhole Reference and Site Name entered                                       |
| <input type="checkbox"/> | Empty Distance entered and level reading checked against a manual measurement |
| <input type="checkbox"/> | Sample rates, send intervals and alarms set to the customer's requirements    |
| <input type="checkbox"/> | NETWORK CONNECT successful (✓) and signal Good or better                      |
| <input type="checkbox"/> | Validate Details checked and NEXT tapped (device active)                      |
| <input type="checkbox"/> | Configuration file downloaded for the site record                             |
| <input type="checkbox"/> | Photographs taken, cover replaced and site left safe                          |
| <input type="checkbox"/> | First dial out confirmed on the monitoring platform                           |

## 8.0 Data Download and Diagnostics

### Data Download

1. **Channel:** select the channels to download.
2. **Start:** select the date and time the download should start from.
3. **End:** select the date and time the download should end at.

### Diagnostics

| Button                  | Function                                                                                                                               |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 'UPDATE FIRMWARE'       | Opens the firmware update page (see Firmware Update).                                                                                  |
| 'NETWORK'               | Opens the Network page to check 4G status, RSRP, RSRQ, SINR and the last attempted and successful dial outs during maintenance visits. |
| 'SET CONFIG USING FILE' | Loads a saved ALARM 2 configuration file from your Android device or Windows PC.                                                       |
| 'DOWNLOAD CONFIG FILE'  | Saves the current ALARM 2 configuration to the downloads folder.                                                                       |
| 'REFRESH REGISTERS'     | Reads the latest register values from the ALARM 2.                                                                                     |
| 'ERASE DATA'            | Permanently deletes all logged data after a YES or NO confirmation. Configuration settings are kept.                                   |

**WARNING** ERASE DATA cannot be undone. Download all logged data from the device before tapping ERASE DATA if you wish to keep it.

**TIP** Use DOWNLOAD CONFIG FILE and SET CONFIG USING FILE to copy a proven configuration to other sites with similar requirements, then update the identifiers and Empty Distance for each site.

## 9.0 Maintenance

The ALARM 2 needs no routine maintenance other than a periodic health check. Carry out the checks below at each site visit.

| Check                     | Action                                                                                                                           |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Visual inspection</b>  | Check the enclosure, skirt, antenna and bracket for damage and for build up of fat, rag or debris. Clean with a damp cloth only. |
| <b>Fixings</b>            | Check the wall socket bolts, locking pin, lanyard and bracket bolts are tight.                                                   |
| <b>Alignment</b>          | Confirm the device is level and over the centre of the flow, particularly after jetting or other works.                          |
| <b>Level accuracy</b>     | Compare the Level Reading with a manual measurement. Update the Empty Distance if the benching or channel has changed.           |
| <b>Battery and signal</b> | Review the heartbeat battery voltage and check RSRP and SINR on the Network page.                                                |
| <b>Data</b>               | Download the logged data for the site record.                                                                                    |
| <b>Firmware</b>           | Update to the latest firmware supplied with the Configurator.                                                                    |

**TIP** Before jetting or other works in the chamber, remove the locking pin and lift the boom arm and ALARM 2 out as one unit. Refit and check the alignment afterwards.

## 10.0 Troubleshooting

| Symptom                                         | Possible cause                                                         | Action                                                                                                                                          |
|-------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ALARM 2 not listed after REFRESH DEVICES</b> | Device asleep, Bluetooth off or out of range                           | Hold the magnet on the activation spot for 5 to 10 seconds, turn on Bluetooth, move closer and refresh again.                                   |
| <b>Disconnected during a firmware update</b>    | Normal restart after the update, or out of range                       | Pair again. Stay close to the device until "Update Complete" is shown.                                                                          |
| <b>NETWORK CONNECT fails (no ✓)</b>             | SIM not active, wrong APN, no coverage or antenna not fitted           | Confirm the ICCID is active with the SIM provider, enter the APN again exactly as supplied, check the antenna and run the signal test.          |
| <b>Connected but no data on the platform</b>    | Wrong server address or port, or send interval Disabled                | Check the server address and port and the send intervals, then check Last Successful Dial Out.                                                  |
| <b>Poor or Very Poor signal</b>                 | Antenna position, nearby metal or a deep chamber                       | Reposition the antenna vertically near the cover opening, or fit an enhanced external 4G LTE antenna.                                           |
| <b>Level readings wrong or unstable</b>         | Incorrect Empty Distance, device not level, obstruction or turbulence  | Measure the Empty Distance again, check the device is level and position it over smooth flow clear of obstructions. Use x5 readings to confirm. |
| <b>Alarms switching on and off</b>              | Threshold close to normal levels, or hysteresis or persistence too low | Review the settings with the customer and increase the hysteresis or persistence.                                                               |
| <b>Battery life shorter than expected</b>       | Short send intervals, poor signal or long periods in alarm             | Lengthen the send intervals, improve the signal and review the alarming send interval.                                                          |
| <b>Movement or fall alert received</b>          | Boom arm or device disturbed                                           | Inspect the installation, check the fixings and realign the device.                                                                             |

If the problem persists, contact Detectronic technical support at [support@detectronic.org](mailto:support@detectronic.org) with the serial number, IMEI and a description of the fault.

## 11.0 Word Dictionary

| Term                           | Meaning                                                                                                                                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>APN (Access Point Name)</b> | The name of the mobile carrier gateway through which the ALARM 2 connects to the monitoring platform. Supplied by the SIM provider.                                                                    |
| <b>Dial Out</b>                | A connection in which the ALARM 2 powers its 4G modem, registers on the mobile network and sends logged data to the monitoring platform.                                                               |
| <b>Empty Distance (ED)</b>     | The distance in mm from the bottom of the sensor skirt to the bottom of the channel. $\text{Water Level (mm)} = \text{Empty Distance} - \text{Sensor Reading}$ .                                       |
| <b>Hysteresis</b>              | The distance the level must fall below the threshold before the alarm clears.                                                                                                                          |
| <b>ICCID</b>                   | Integrated Circuit Card Identifier. The unique serial number of the SIM.                                                                                                                               |
| <b>IMEI</b>                    | International Mobile Equipment Identity. The unique 15 digit identifier of the 4G modem.                                                                                                               |
| <b>LIDoTT Configurator</b>     | Android and Windows application used to configure Detectronic level and flow monitoring products.                                                                                                      |
| <b>LTE Cat 1</b>               | A 4G LTE device category (UE Category 1) offering up to 10 Mbit/s downlink and 5 Mbit/s uplink on standard 4G networks.                                                                                |
| <b>Persistence</b>             | The number of further consecutive readings above the threshold, after the first, needed to raise an alarm. With persistence 1, two readings in a row are needed; with persistence 2, three are needed. |
| <b>RSRP</b>                    | Reference Signal Received Power, in dBm. The main measure of 4G signal strength; values closer to zero are stronger.                                                                                   |
| <b>RSRQ</b>                    | Reference Signal Received Quality, in dB. A measure of signal quality that accounts for interference and cell load.                                                                                    |
| <b>SINR</b>                    | Signal to Interference plus Noise Ratio, in dB. Higher values indicate a cleaner signal.                                                                                                               |
| <b>STANAG 4694 NATO</b>        | The NATO Accessory Rail standard, used to mount the laser alignment tool.                                                                                                                              |
| <b>SWMS</b>                    | Safe Work Method Statement.                                                                                                                                                                            |
| <b>Threshold</b>               | The water level that must be reached for an alarm to be raised.                                                                                                                                        |

## 12.0 End of Life Management

### Our Commitment to ESG

At Detectronic, our commitment to Environmental, Social and Governance (ESG) standards is fully integrated within the broader ESG strategy of our parent company, Adler & Allan. We actively support sustainable practices across our operations, with a particular focus on responsible device end of life management.

### Device End of Life Management

- **Sustainable design:** reducing waste by creating products that are built to last.
- **Recyclable materials:** using recyclable plastics for the outer casings of our devices.
- **Refurbishment and reuse:** extending device life through effective repair and reuse.
- **Responsible recycling:** partnering with certified recycling providers for safe, compliant and eco friendly disposal of electronic waste.

**NOTE** Before a device is returned for disposal, ask the SIM provider to deactivate the SIM so that no further data charges are incurred.

Detectronic has established a service with our partners to responsibly dispose of our products at end of life. To find out more, contact your account manager or [support@detectronic.org](mailto:support@detectronic.org).

## 13.0 Contact Detectronic

| Contact           | Details                                                                               |
|-------------------|---------------------------------------------------------------------------------------|
| Address           | Ground Floor Suite, 16 Lindred Road, Brierfield, Nelson, Lancashire, BB9 5SR, England |
| Telephone         | +44 1282 449124                                                                       |
| Web               | <a href="http://www.detectronic.org">www.detectronic.org</a>                          |
| Technical support | <a href="mailto:support@detectronic.org">support@detectronic.org</a>                  |

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